



PRISM

Plasma **R**esearch **I**ntegrated **S**ystem for **M**ulti-diagnostics

v2.3.0

2026.03.24

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Overview

What is PRISM

PRISM is a software platform for viewing and analyzing multi-diagnostic data from the **KSTAR tokamak** in a [unified interface](#).

It directly connects to the MDS+ database to load experimental data, providing various visualizations from profile analysis to spectrograms and imaging analysis.

How to Run: Execute `$ prism` command on nkstar server

Key Numbers

12

Diagnostics

12

Tabs

5

EFIT Trees

2

Themes (Dark/Light)

Supported Diagnostics

DIAGNOSTIC	FULL NAME	CHANNELS	PARAMETERS	TABS
CES	Charge Exchange Spectroscopy	32	T_i, v_T	Profile Time Trace
XICS	X-ray Imaging Crystal Spectrometer	1	T_{i0}, v_{T0}	Profile Time Trace
TS	Thomson Scattering	31	T_e, n_e	Profile Time Trace
ECE	Electron Cyclotron Emission	76	T_e	Profile Time Trace Spectrogram
MSE	Motional Stark Effect	25	γ, q, j	Profile Time Trace
BES	Beam Emission Spectroscopy	4x16	Intensity	Spectrogram
TCI	Two-Color Interferometer	5	n_e (line)	Spectrogram Time Trace
ECEI	ECE Imaging	24x8x3	T_e fluct.	Spectrogram
MC	Mirnov Coil	20T/22P	dB/dt	Spectrogram
TV	In-Vessel Inspection System TV	-	Video	Imaging
IRVB	Infra-Red Video Bolometer	-	P_{rad}	Imaging
Neutron	Fusion Neutron Diagnostics	3	Intensity	Time Trace

Key Features



12 Diagnostics

CES, TS, ECE, MSE, BES, TCI, ECEI, MC, TV, IRVB, Neutron in one UI



Dark/Light Themes

Toggle via bottom toolbar for all widgets and plots



EFIT Mapping

5 EFIT trees for R → flux coordinate transformation



Data Export

Preview spreadsheet then save as .csv or .npz or .png or .svg



Plot Options

Colormap, font sizes per tab.
Settings persist across sessions



Channel Toggle

Show/high individual channels in Profile to highlight specific data



On-demand Loading

Tabs initialize only when selected for fast startup



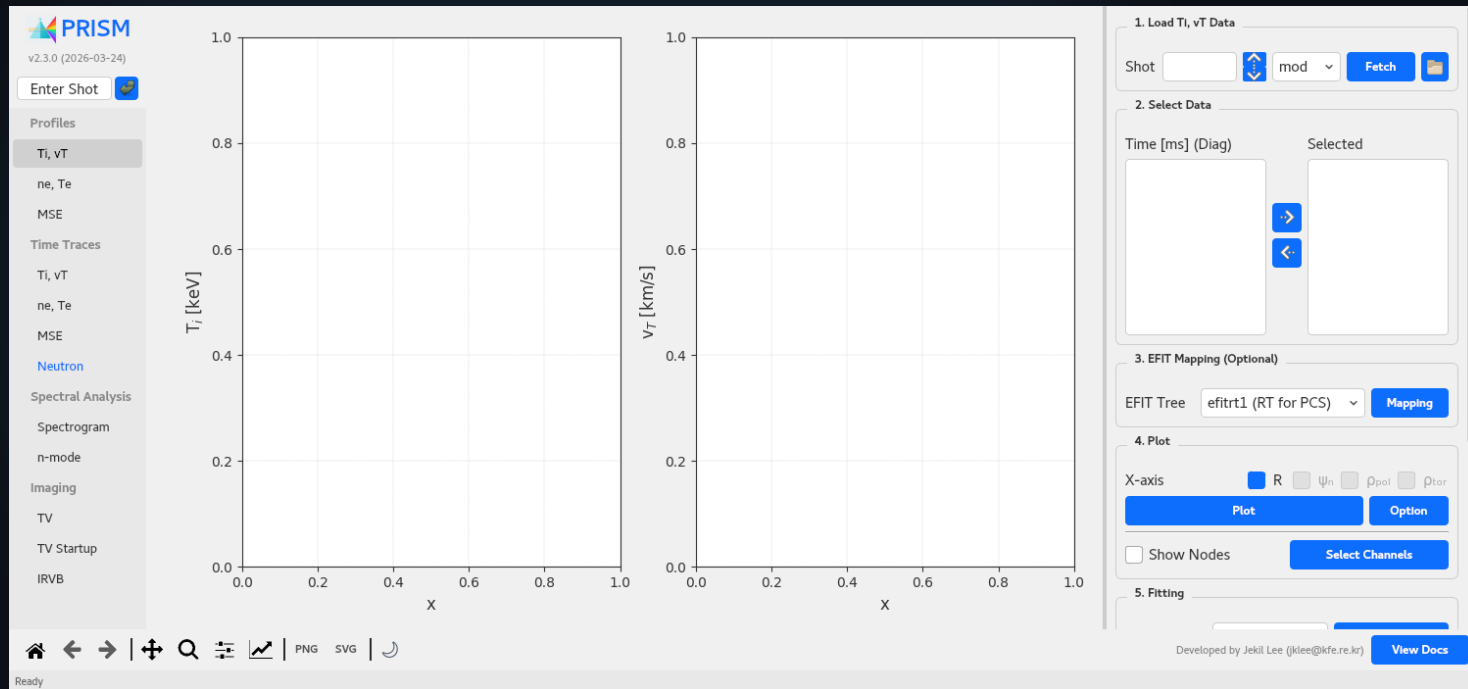
What's New

Auto-popup on update shows changelog

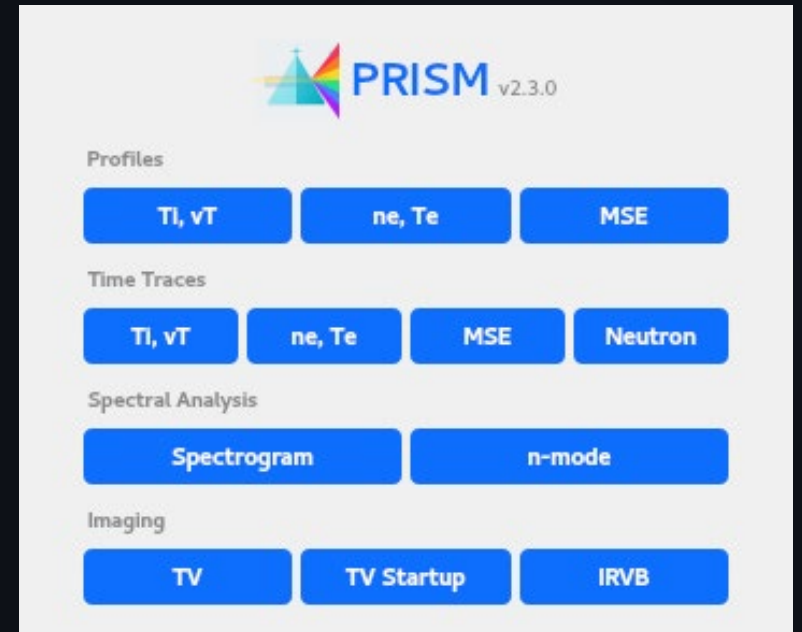
User Interface

UI Structure

Full Mode `$ prism`



Selector Mode `$ prism -s`



● **Sidebar** 4 categorizes, 12 tabs

● **Control Panel** Fetch/Select/Mapping/Fitting/Save Data

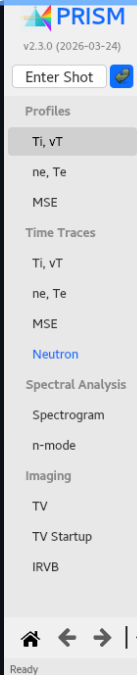
● **Plot Area** matplotlib interactive

● **Bottom Toolbar** Pan/Zoom, Axis Control, Save Figure, Docs

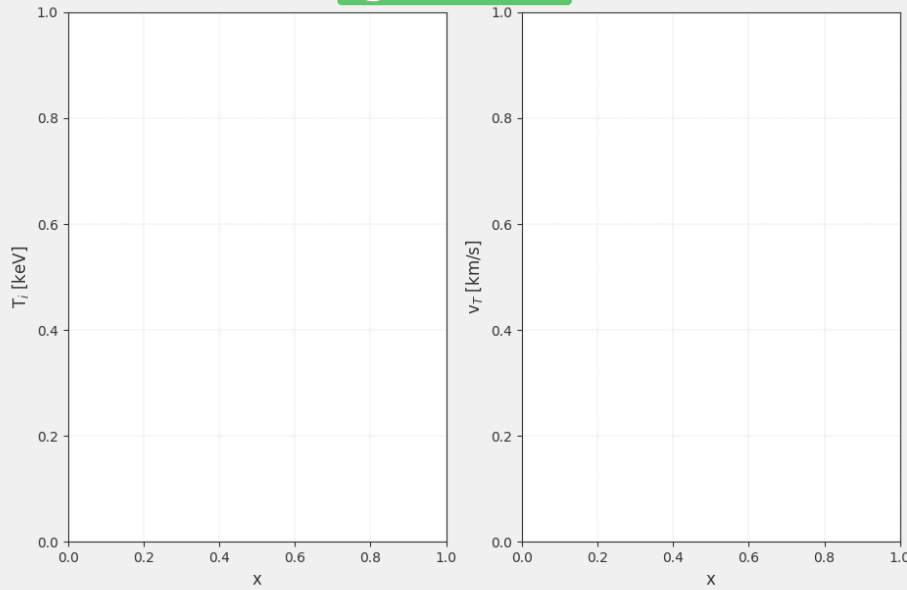
User Interface

GUI Function Guide

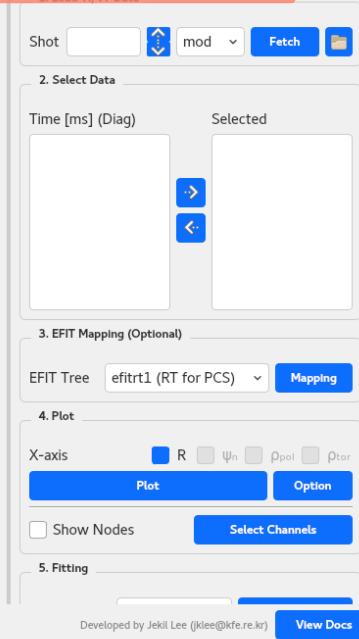
① Sidebar



② Plot Area



③ Control Panel



④ Bottom Toolbar



① Sidebar

- **Profile:** T_i , v_T / n_e , T_e / MSE profiles
- **Time Traces:** Time-series plots
- **Spectral Analysis:** Spectrogram, n-mode
- **Imaging:** TV, Startup, IRVB

③ Control Panel

1. **Load Data:** Enter shot number, Fetch button
2. **Select Data:** Time/channel selection
3. **EFIT Mapping:** Execute plot, Options (font size, etc...)
4. **Plot Mapping:** X-axis select, Plot, Options
5. **Fitting:** Function select, Fit, Option
6. **Save Data:** Preview and CSV export

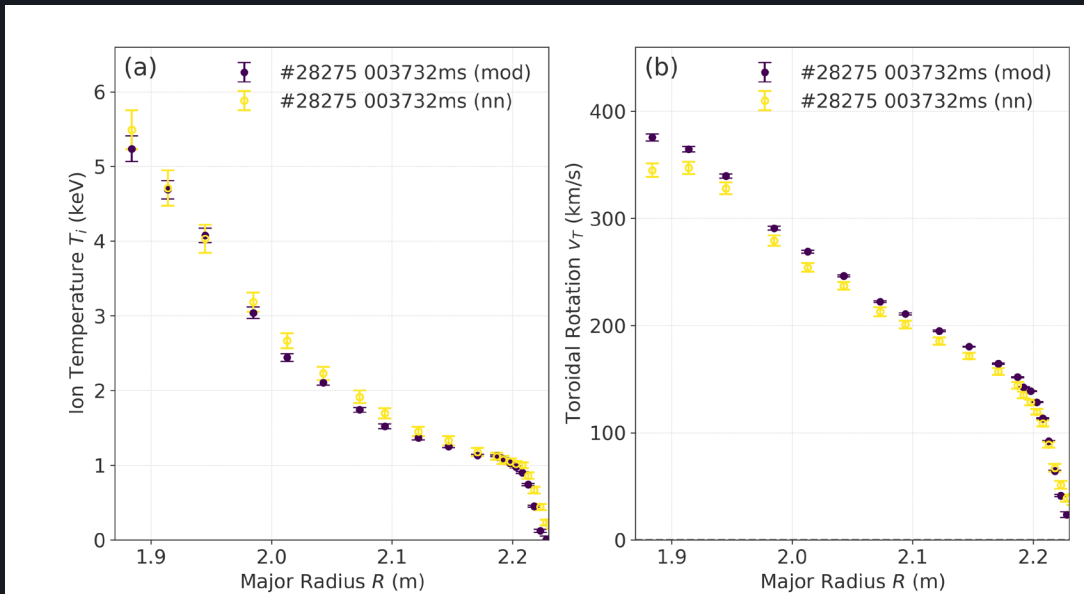
④ Bottom Toolbar

- | | |
|------------------------------|-------------------------------------|
| Home: Reset view | Back/Forward: Navigate views |
| Pan: Drag to move | Zoom: Zoom to region |
| Axes: Set axis range | Margin: Set margin range |
| Save: as *.png, *.svg | Theme: Dark/Light |

Profiles

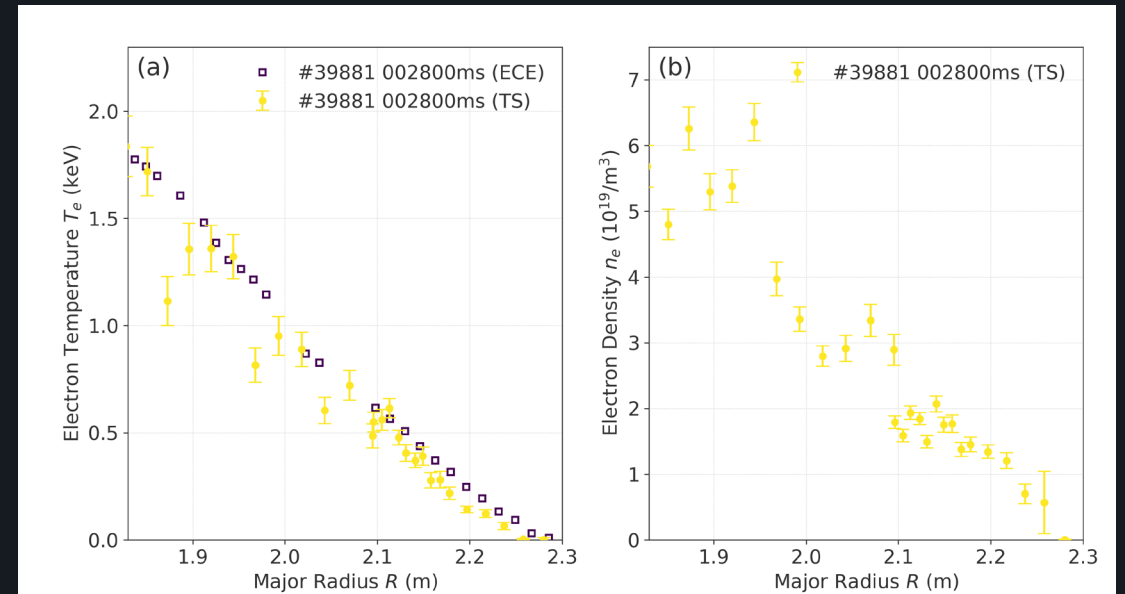
T_i , v_T Profile & n_e , T_e Profile

Profile T_i , v_T Ion Temperature & Toroidal Rotation



- **CES (*mod/nn) + XICS overlay**
* mod: Beam Modulation / nn: Neural Network [[J. K. Lee FED \(2026\)](#)]

Profile n_e , T_e Electron Density & Temperature

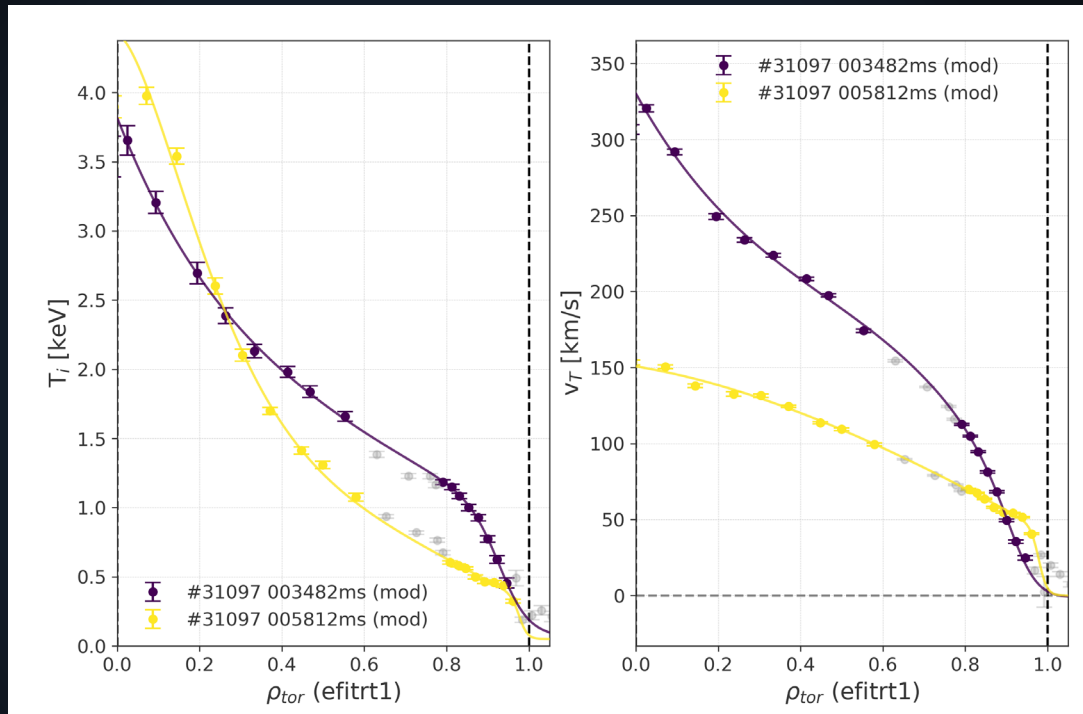


- **TS + ECE overlay**
- **Auto channel position**

Common Features EFIT coordinate mapping (5 trees) · 9 colormaps · Font size settings · Data preview/export (.csv, .npz)

Profiles

T_i , v_T & n_e , T_e Profile Fitting



Available Functions

- mtanh** — Modified hyperbolic tangent (8 parameters)
- ptanh** — Polynomial $\times$ tanh (7 parameters)
- EPED** — Edge Pedestal model (6 parameters)
- Spline** — Cubic spline interpolation
- RBF** — Radial Basis Function (adjustable bases)
- GPR** — Gaussian Process Regression ($\pm 2\sigma$ bands)

Features

- Option dialog with editable parameter table (Value, Min, Max, Fix)
- Pedestal analysis: HEIGHT, FOOT, WIDTH, LOC
- Double-click a data point to exclude/include it from fitting
- Preview & Save: export fit results with parameters

```
[Fitting] 031097 003482 (mod) Ti (mtanh): chi2=0.4476
[Fitting] Param Value Error Min Max Fixed
[Fitting] a1 0.0500 0.6502 0.05 0.3
[Fitting] a2 0.9589 0.1028 0 inf
[Fitting] a3 0.1447 0.0745 0.04 0.15
[Fitting] a4 0.9352 0.0369 0.85 1.02
[Fitting] a5 0.2243 0.2439 0 2
[Fitting] a6 3.8159 0.2790 0 10
[Fitting] a7 0.0000 0.1858 0 0.5
[Fitting] a8 1.0100 0.4664 1.01 10
[Fitting] PED HEIGHT = 0.9589 +/- 0.1028 [keV]
[Fitting] PED FOOT = 0.0500 +/- 0.6502 [keV]
[Fitting] PED WIDTH = 0.1447 +/- 0.0745 [rho_tor] = 3.15 [cm]
[Fitting] PED LOC = 0.9352 +/- 0.0369 [rho_tor] = 2.2207 [m]
```


MSE Profile

Measured Parameter

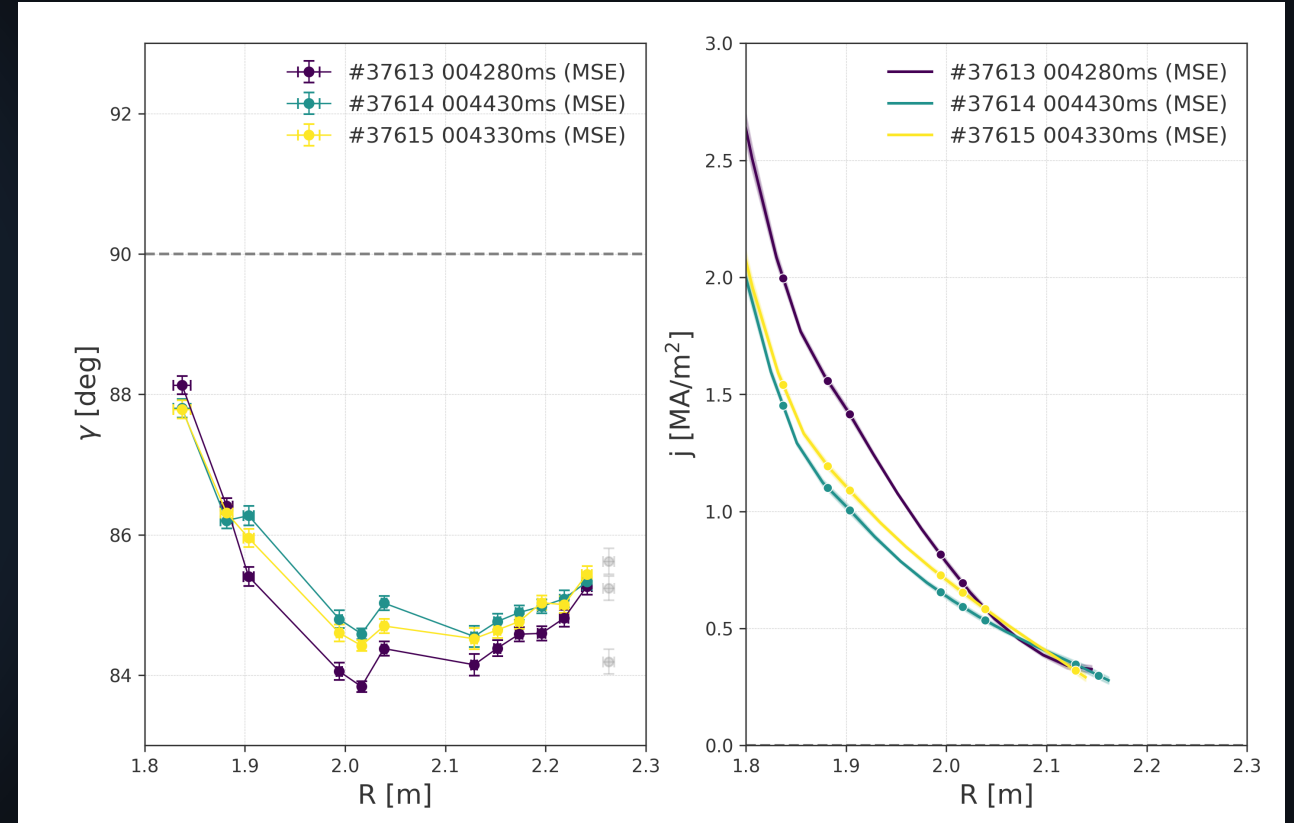
- Pitch Angle ($\tan \gamma$): MSE raw data (25 channels)

Derived Parameters [\[C. C. Petty NF \(2002\)\]](#)

- **q Profile:** Safety factor
- **j Profile:** Current density

Special Features

- Auto NB source detection (NB-1A / 1B / 1C)



Time Traces

Time Trace Tabs

T_i , v_T Time Trace

CES / XICS

- CES channel-wise T_i , v_T evolution
- XICS overlay available

n_e , T_e Time Trace

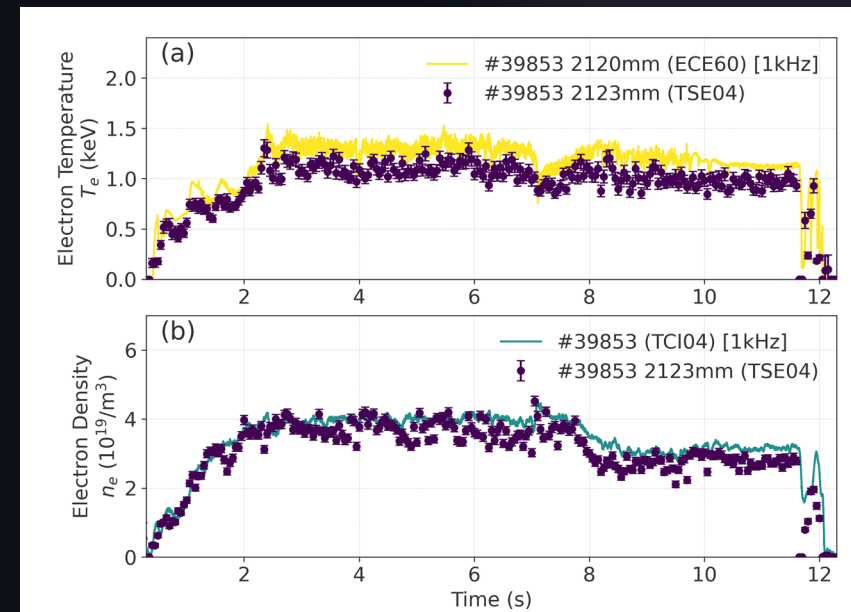
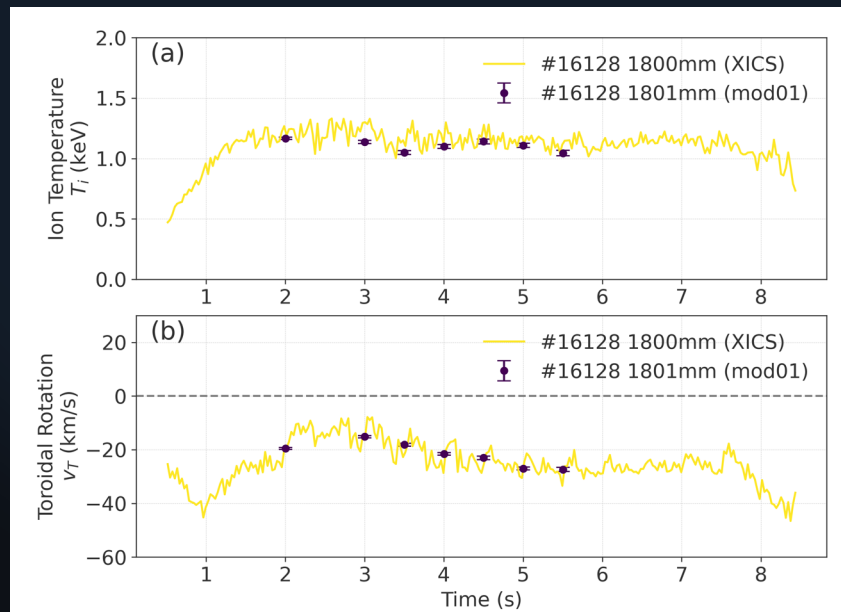
Thomson / ECE / TCI

- TS/ECE channel-wise T_e , n_e
- Diagnostic selection available

MSE Time Trace

MSE

- $\tan \gamma$, q , j time evolution
- q/j toggle radio buttons



Neutron Time Trace

Fusion Neutron Diagnostics (near J-port)

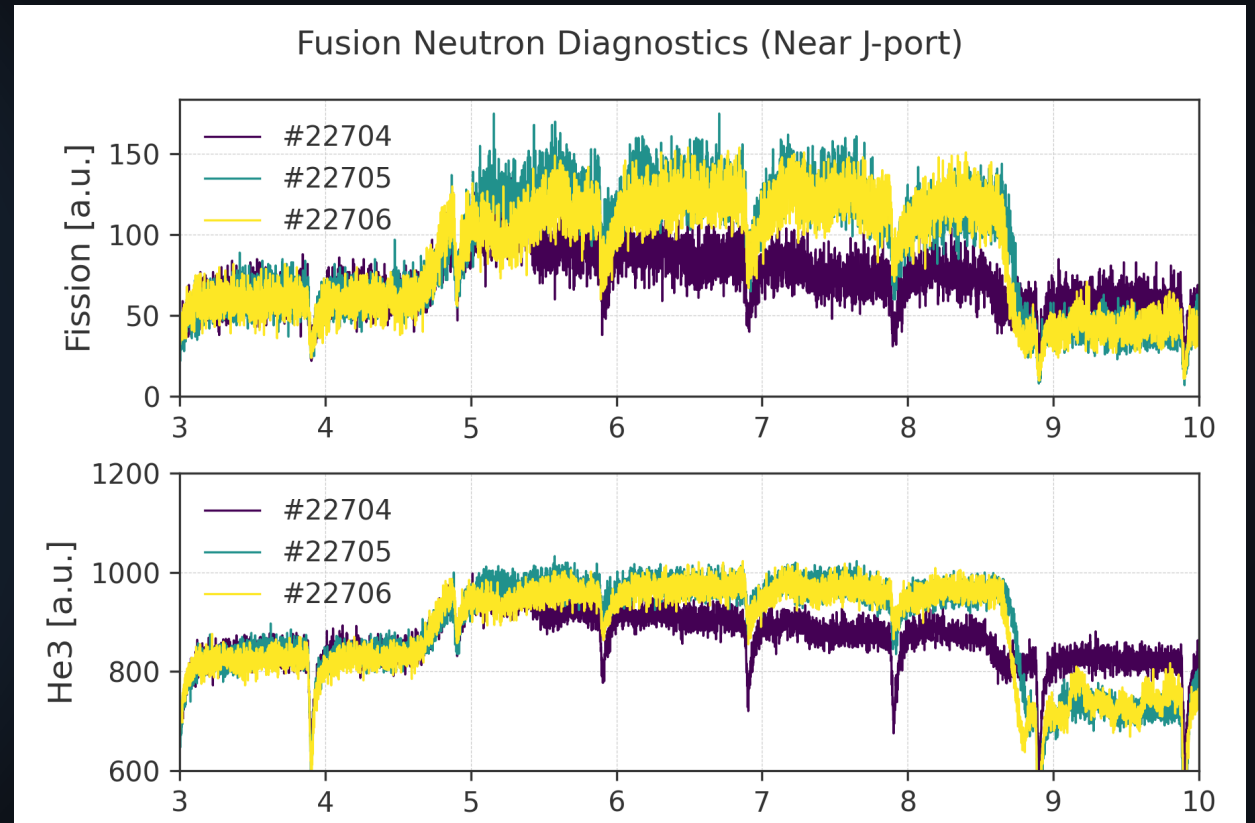
Detectors (3×1 subplot layout)

- **Fission Chamber** — Thermal neutron detection
- **He-3 Counter** — Proportional counter
- **Diamond Detector** — Fast neutron measurement

Key Features

- Multi-shot overplot comparison
- Shared x-axis across 3 subplots
- Plot Options (color, font sizes)
- Data export (.csv, .npz)

* Related study: Magnetic Breaking [\[K. Kim NF \(2023\)\]](#)



Spectrogram

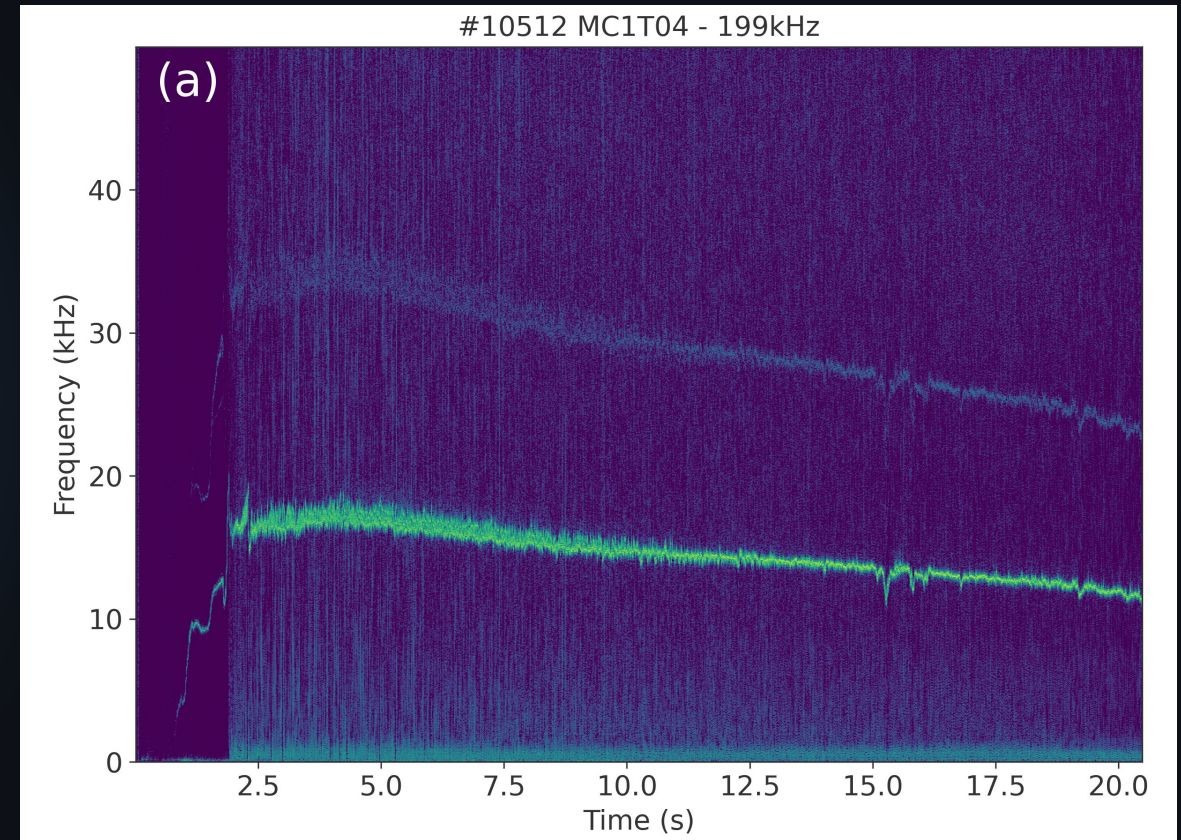
Spectrogram

Supported Diagnostics

ECE	Electron Cyclotron Emission	76 ch
MSE	Motional Stark Effect	25
BES	Beam Emission Spectroscopy	4x16 ch
TCI	Two-Color Interferometer	5 ch
ECEI	ECE Imaging	24x8x3 ch
MC	Mirnov Coil	20T/22P ch

Configuration Parameters

- NFFT, dynamic range settings
- Time range & frequency range settings
- NPZ data export



Spectrogram

n-Mode Spectrum

Data Source Measured Parameter

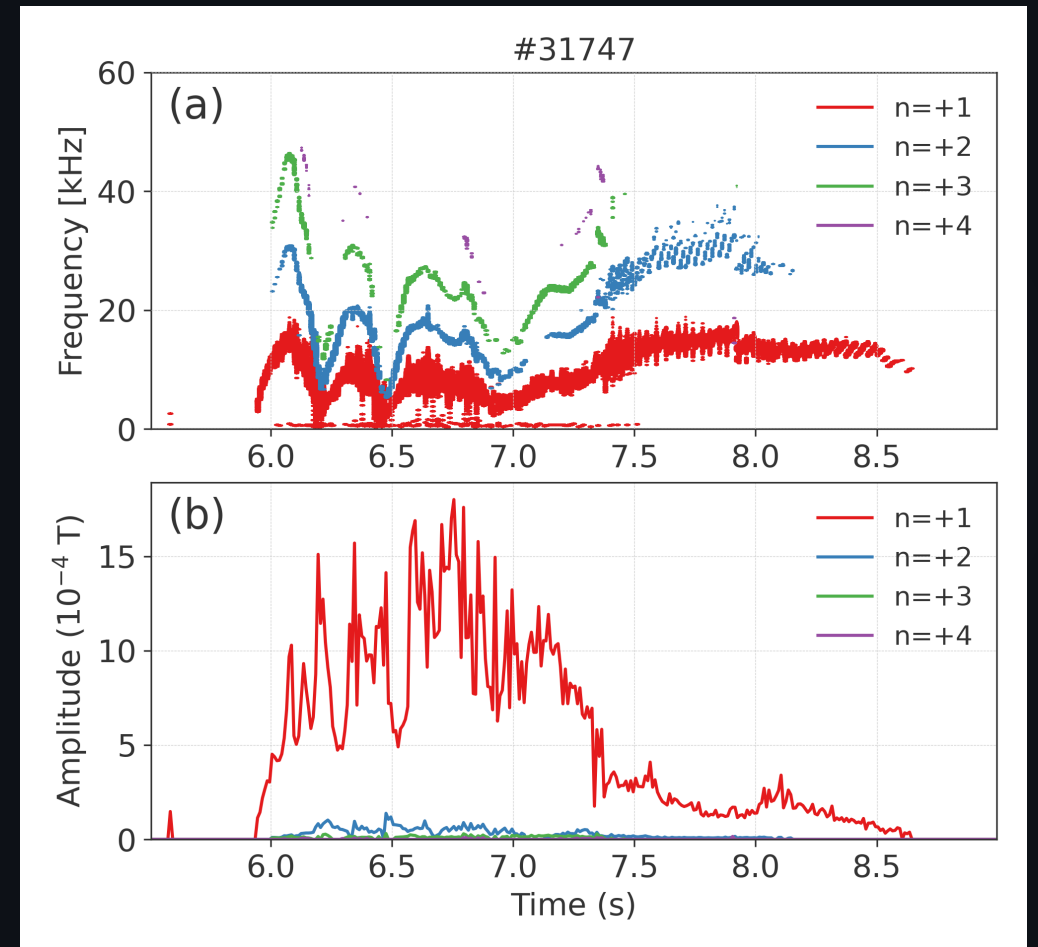
FFT analysis of dB/dt data from MC1T (toroidal Mirnov coils) to calculate amplitude spectrum for toroidal mode numbers

Analysis Options

- Mode Sign, Integrate, Detrend, Plot Type, Color

Output

- Top: Time-frequency map
- Bottom: Mode amplitude
- NPZ export supported



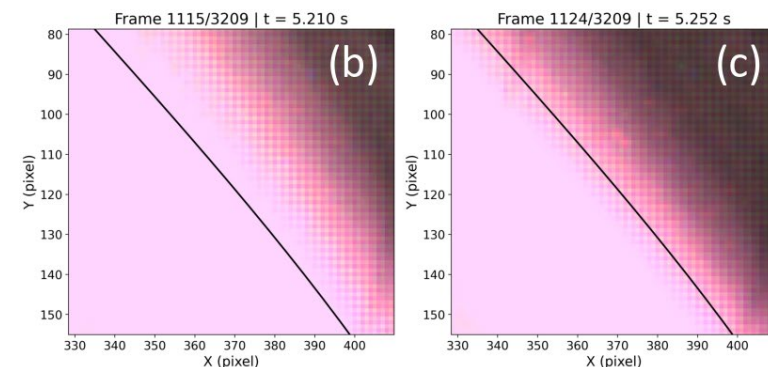
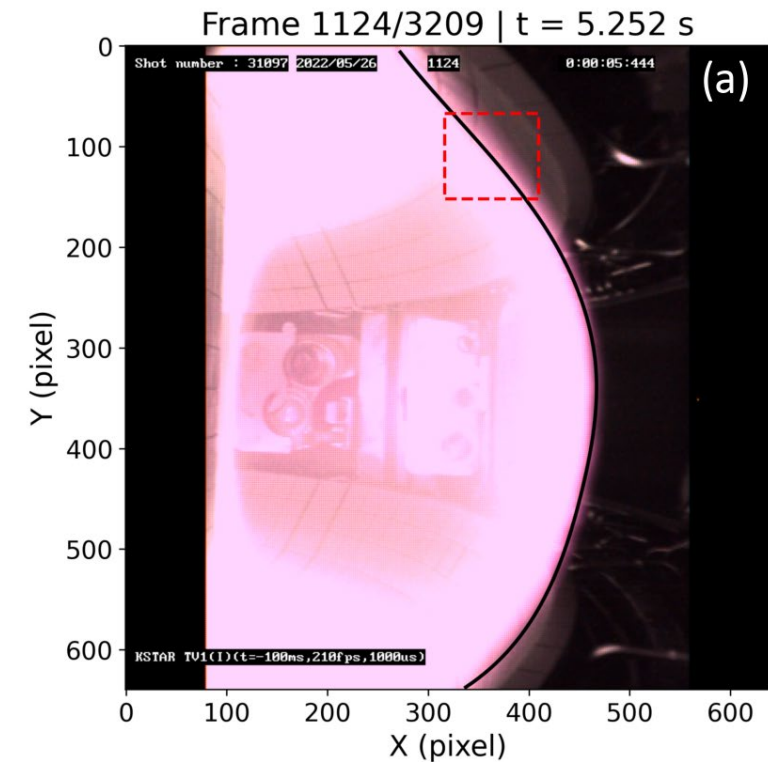
Imaging

TV

Key Features

- Auto-load images by shot number
- Frame slider / prev-next / direct input / mouse wheel
- Auto-display time (seconds) for each frame
- Compare mode: TV01 + TV02 side-by-side
- Play/Pause/Stop playback control, FPS adjustment
- Draw Mode: Spline interpolation curve drawing

* Related study: Edge kink-like mode observation [[J. K. Lee PoP \(2025\)](#)]

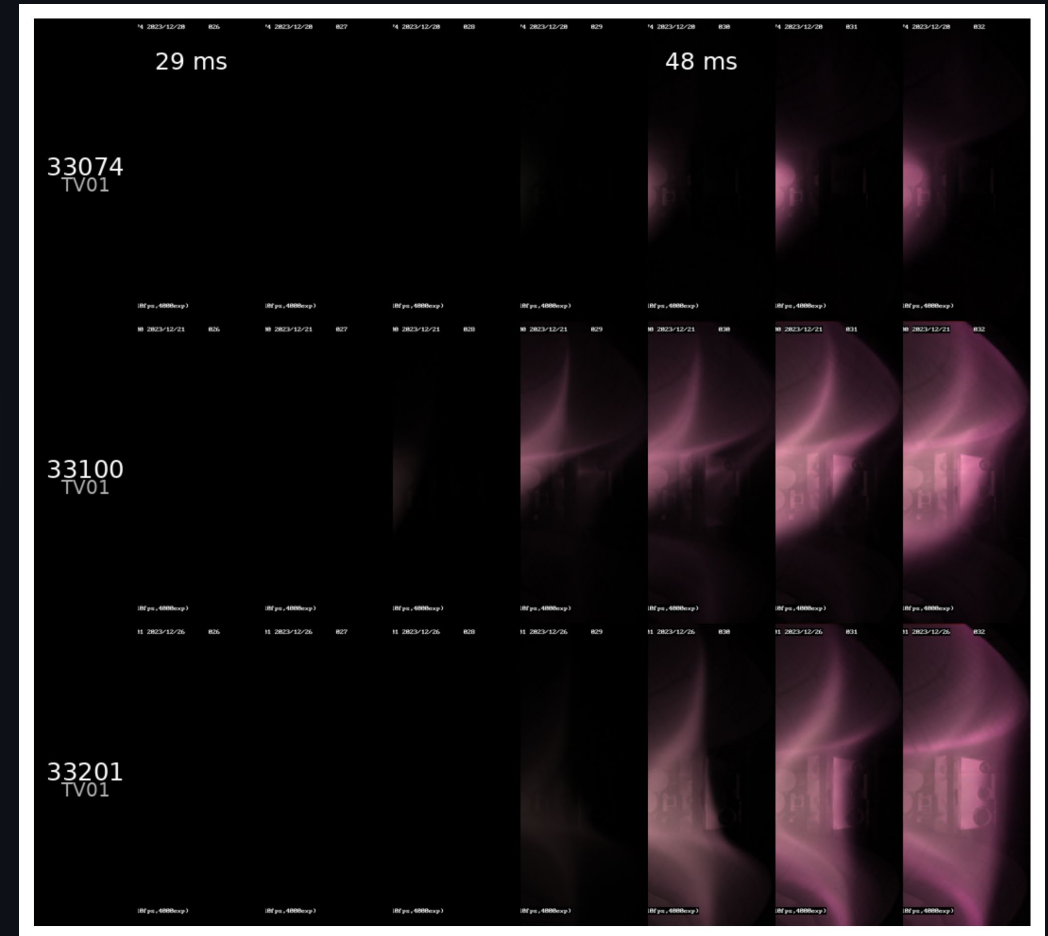


TV Startup

Key Features

- Compare TV images from multiple shots in **grid layout**
- TV01/TV02 **free mixing** comparison
- Set frame range or time range

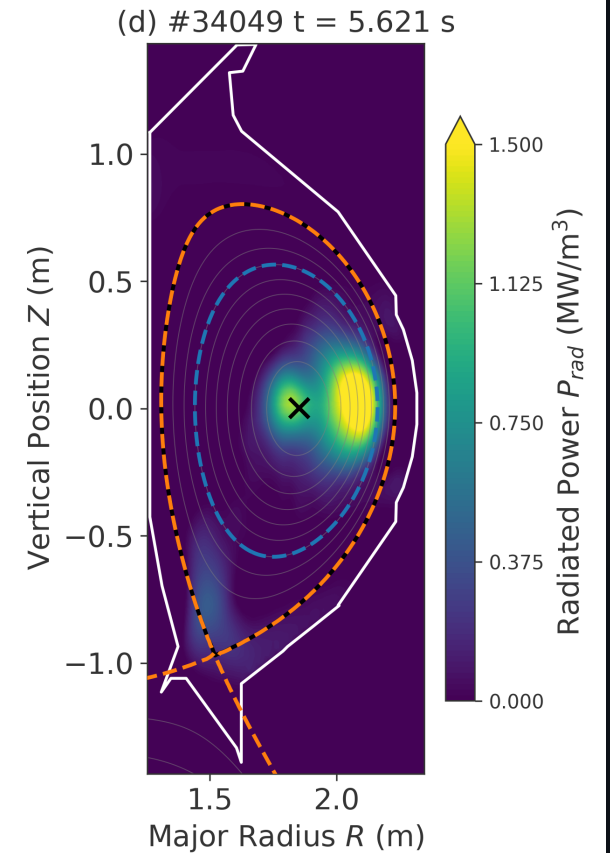
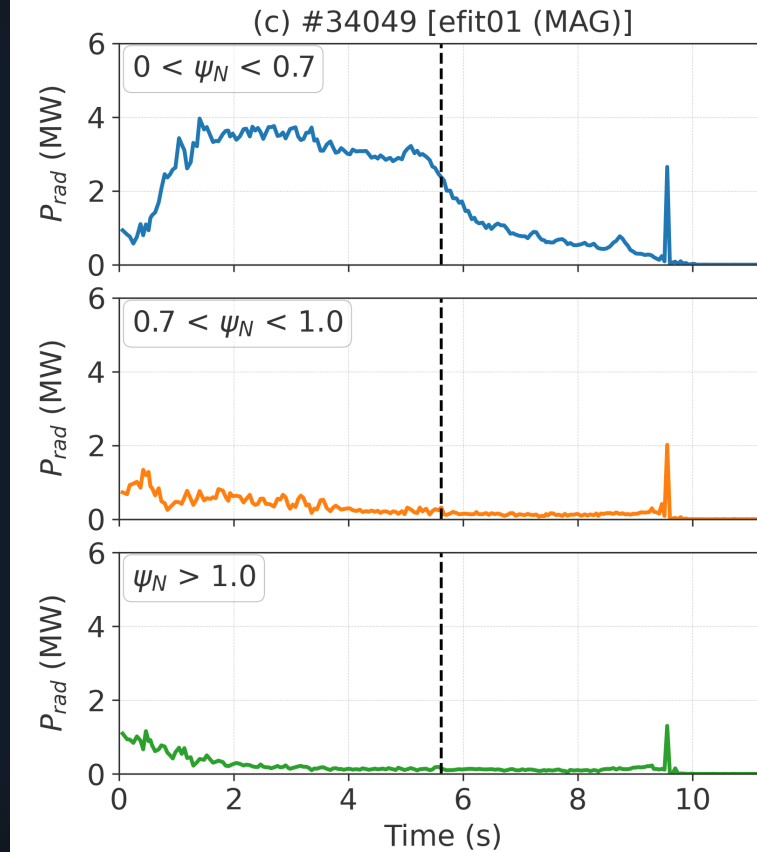
* Related study: Startup Scenario Optimization [[J. Lee FED \(2024\)](#)]



IRVB

Key Features

- 2D P_{rad} heatmap
- EFIT overlay
- User-defined flux region analysis
- Region-wise P_{rad} traces
- Frame navigation



Common Features

Data Export & EFIT Mapping



Data Export

Preview plotted data in spreadsheet format before saving.

.csv Format

Tab-delimited CSV file. Open directly in spreadsheet

.npz Format

NumPy compressed binary + example script auto-generated

Data Copy

Copy data to clipboard from preview into Excel, etc.



EFIT Coordinate Mapping

Compute EFIT mapping, then select flux coordinate via X-axis radio buttons in Plot section.

Coordinate Types

ψ_N

ρ_t

ρ_p

EFIT Trees (5 types)

efitrt1

Magnetic constraints for real-time

efitrt2

Magnetic and MSE constraints for real-time

efit01

Magnetic constraints

efit02

Magnetic and MSE constraints

efit04

Drift-corrected magnetic constraints



Theme

Dark/Light themes available



Plot Options

Colormaps, font sizes



What's New

Auto-popup changelog after updates



Thank you for using **PRISM!**

Questions, suggestions, and bug reports are always welcome.

 : jkleee@kfe.re.kr